Seismic Wave Processing using Python

A Report on Digital Signal Processing for Earthquake Parameter Estimation on Python

1. Introduction

Earthquakes are one of the most destructive natural events on earth. When an earthquake happens, it releases energy in the form of seismic waves that travel through the ground. These waves can be detected using a device called a seismograph. By studying these waves carefully, scientists can figure out where the earthquake happened, how far away it was, and how strong it was.

There are two main types of seismic waves that are important for analysis. The first is called the P-wave, which stands for Primary wave. It is the faster of the two and arrives at the seismograph first. The second is called the S-wave, or Secondary wave, and it arrives a bit later. The time gap between the P and S wave arrivals is very useful because it tells us how far the earthquake was from the station.

In this project, a Python program was written to simulate this whole process. The code generates a fake seismic signal (similar to what a real seismograph would record), filters out unwanted noise from it, detects the P and S waves automatically, and then calculates the distance to the earthquake and its magnitude on the Richter scale. The entire process from signal creation to final estimation mirrors what real seismological software does, just in a simpler and more understandable way.

The main tools used in this simulation are NumPy for signal generation and mathematics, SciPy for digital filtering, and Matplotlib for plotting the results. The detection algorithm used is called STA/LTA, which is a standard method in seismology and is explained in detail in the Methods section.

2. Principles

2.1 Seismic Waves

When the ground breaks or slips during an earthquake, it sends out vibrations in all directions. These are called seismic waves. As mentioned before, two of the most important ones for earthquake detection are P-waves and S-waves.

P-waves travel faster (roughly 6 km/s in typical crustal rock) and cause the ground to compress and expand in the same direction the wave is moving. S-waves are slower (around 3.5 km/s) and shake the ground sideways. Because they arrive later and shake the ground more violently, they cause more damage. The time difference between when the P-wave arrives and when the S-wave arrives, written as the S-P time, is directly related to how far away the earthquake was.

2.2 Analog vs. Digital Signals

In the real world, ground motion is continuous and smooth. This is called an analog signal. But computers can only work with numbers, so the continuous signal needs to be sampled at regular time intervals. This process is called Analog to Digital Conversion (ADC). The number of samples taken per second is called the sampling frequency or sampling rate, written as fs.

In this code, an analog-style signal is first generated at a very high sampling rate (5000 Hz) to look smooth and continuous. Then a separate digital version is created at a much lower rate (50 Hz) to simulate what a real digital seismograph would record. The Nyquist theorem tells us that to correctly capture a signal, the sampling frequency must be at least twice the highest frequency present in the signal. Since most earthquake signals have energy below 10 Hz, sampling at 50 Hz is more than enough.

2.3 Digital Filtering

Real seismograph recordings always contain noise. This noise comes from traffic, wind, machines, and other background vibrations. To get a cleaner signal, a bandpass filter is applied. A bandpass filter allows only a specific range of frequencies to pass through and blocks everything else. In this program, a Butterworth bandpass filter is used with a passband from 0.5 Hz to 10 Hz. The Butterworth design is popular because it has a flat response in the passband, meaning it doesn't distort the frequencies it lets through. The filter is applied using a method called filtfilt, which applies the filter twice (once forward, once backward) to avoid any phase shift. This makes sure the detected wave arrival times are accurate

3. Methods

3.1 Synthetic Signal Generation

Since real seismograph data wasn't available, a synthetic (fake but realistic) seismic signal was created mathematically. The signal has three parts: random background noise, a P-wave, and an S-wave.

The noise is generated using a Gaussian random number function. The P-wave is modeled as a sine wave with a frequency of 4 Hz that starts at time t = 15 seconds and decays quickly because real P-waves don't last long. The S-wave is also a sine wave but at a lower frequency of 2 Hz, starts at t = 23 seconds, has a much bigger amplitude (300 vs 40), and decays more slowly. This is consistent with how real seismic waves behave. The exponential decay term controls how quickly the wave fades away after arrival.

The signal is generated twice: once at 5000 Hz to represent the continuous analog waveform (for display purposes), and once at 50 Hz to represent the digitized signal that will be processed.

3.2 Butterworth Bandpass Filter

After generating the digital signal, a 4th order Butterworth bandpass filter is applied to clean it up. The cutoff frequencies are set to 0.5 Hz and 10 Hz. These were chosen because the useful seismic content (P and S waves) falls in this range, while noise from the environment tends to be outside it.

The filter coefficients (b and a) are calculated using the butter() function from SciPy. The filtfilt() function then applies the filter in a zero-phase manner. This step significantly improves the clarity of the wave arrivals and makes detection easier.

3.3 STA/LTA Detection Algorithm

The STA/LTA method is one of the oldest and most widely used techniques in seismology for detecting earthquake wave arrivals. STA stands for Short-Term Average and LTA stands for Long-Term Average.

The idea is simple: you take the absolute value of the filtered signal and calculate two running averages over different window lengths. The STA uses a short window of 1 second (50 samples at 50 Hz) and responds quickly to sudden changes in signal energy. The LTA uses a longer window of 10 seconds (500 samples) and gives a baseline of what the signal normally looks like.

The ratio of STA to LTA is then computed at every point in time. When a wave arrives, the short-term energy shoots up suddenly while the long-term average stays low. This causes the ratio to spike. When the ratio crosses a threshold value (set to 3 in this code), it signals a wave arrival. The first time the ratio exceeds 3 is taken as the P-wave arrival. Then, to find the S-wave, the algorithm skips ahead by 3 seconds (to avoid re-detecting the P-wave) and looks for the next time the ratio exceeds 3.

3.4 Distance Estimation

Once the P and S wave arrival times are detected, the distance to the earthquake can be estimated. The formula used is based on the fact that both waves start from the same point at the same time but travel at different speeds.

If Vp is the speed of the P-wave (6 km/s) and Vs is the speed of the S-wave (3.5 km/s), and delta_t is the S-P time difference in seconds, then the distance D is:

$$D = (Vp \times Vs) / (Vp - Vs) \times delta_t$$

This gives the straight-line distance from the seismograph station to the earthquake epicenter in kilometers. In real seismology, at least three stations are needed to pinpoint the exact location, but a single station can tell you the distance (and you can draw a circle of that radius around the station).

3.5 Richter Magnitude Estimation

The Richter magnitude scale is a logarithmic scale used to measure earthquake strength. In this program, the local magnitude (ML) is estimated using a simplified version of the original Richter formula:

$$ML = \log10(A) - (1.11 \times \log10(D) + 0.00189 \times D - 2.09)$$

Here, A is the maximum amplitude of the signal in the 5 seconds following the S-wave arrival (in micrometers), and D is the estimated distance in kilometers. The terms in brackets act as a correction factor that accounts for how seismic waves naturally lose energy as they travel farther from the source. The result gives an approximate magnitude on the Richter scale.

4. Pseudocode

The following pseudocode summarizes the logical flow of the program step by step:

```
BEGIN

  // Step 1: Set parameters
  SET duration = 60 seconds
  SET fs_analog = 5000 Hz  (for display only)
  SET fs_digital = 50 Hz   (for processing)
  SET P-wave true time = 15 s, S-wave true time = 23 s

  // Step 2: Generate synthetic seismic signal
  FUNCTION generate_seismic(time_array):
    noise = random Gaussian noise
```

```
    p_wave = 40 * sin(2*pi*4*(t - 15)) * exp(-0.9*(t - 15))
for t >= 15
    s_wave = 300 * sin(2*pi*2*(t - 23)) * exp(-0.35*(t - 23))
for t >= 23
    RETURN noise + p_wave + s_wave
  END FUNCTION

  // Step 3: Apply bandpass filter
  DESIGN Butterworth filter, order=4, passband=[0.5 Hz, 10
Hz]
  APPLY zero-phase filter (filtfilt) to digital signal

  // Step 4: STA/LTA computation
  STA_window = 1 second = 50 samples
  LTA_window = 10 seconds = 500 samples
  FOR each time step:
    STA = mean(|signal| over last STA_window samples)
    LTA = mean(|signal| over last LTA_window samples)
    ratio = STA / LTA
  END FOR

  // Step 5: Detect P-wave
  p_detect_index = first index where ratio > 3
  p_detect_time = time at p_detect_index

  // Step 6: Detect S-wave
  SKIP 3 seconds ahead from p_detect_index
  s_detect_index = next index where ratio > 3
  s_detect_time = time at s_detect_index

  // Step 7: Estimate distance
  delta_t = s_detect_time - p_detect_time
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  distance = (Vp * Vs) / (Vp - Vs) * delta_t

  // Step 8: Estimate magnitude
  A = max amplitude in 5s window after S-wave arrival
  ML = log10(A) - (1.11*log10(distance) + 0.00189*distance -
2.09)

  // Step 9: Plot all results
  PLOT continuous analog waveform with true P and S times
  PLOT filtered digital signal with detected P and S times
  PLOT STA/LTA ratio with threshold line

END
```

5. Results

When the program is run, the following types of outputs are produced. Note that because the noise in the signal is randomly generated each time, the exact numbers may vary slightly between runs. The values mentioned here are representative of a typical run.

5.1 Console Output

The program prints three key results to the console:

- Detected P-wave time: approximately 15 to 16 seconds (should be close to the true value of 15 s)
- Detected S-wave time: approximately 23 to 24 seconds (close to the true value of 23 s)
- Estimated distance to earthquake: typically around 25 to 30 km (based on the S-P time gap of about 8 seconds)
- Estimated Richter Magnitude: typically in the range of 3 to 5 depending on amplitude and distance

The detected times may not match the true values exactly because the STA/LTA method detects when the signal energy rises above the threshold, which can be a tiny bit after the wave actually arrives. But in practice it is quite close, usually within half a second or so.

5.2 Plots

Three plots are generated by the program:

Plot 1 - Continuous Seismic Waveform: This shows what the signal looks like as a continuous, smooth waveform (like what an old analog seismograph would record). Vertical dashed lines mark the true P and S wave arrival times. The P-wave is visible as a smaller burst starting at 15 seconds, and the S-wave is visible as a much larger and more dramatic burst starting at 23 seconds. The difference in amplitude between the two is clear just by looking at the graph.

Plot 2 - Filtered Digital Signal: This shows the sampled and filtered version of the signal as a stem plot, where each sample is shown as a vertical stick. The green dashed line marks the detected P-wave and the red dashed line marks the detected S-wave. The filtering has removed a lot of the background noise, making the wave arrivals much cleaner and easier to see compared to the raw signal.

Plot 3 - STA/LTA Ratio: This plot shows how the STA/LTA ratio changes over time. Before any wave arrives, the ratio stays flat and low (close to 1). When the P-wave hits, there is a spike that crosses the threshold line. After a brief drop, another bigger spike occurs when the S-wave arrives. This plot makes it very easy to understand why and how the detection algorithm works.

5.3 Accuracy Observations

The detection accuracy is generally good for this type of synthetic signal. The P-wave detection tends to be slightly delayed because the P-wave amplitude (40 units) is relatively small compared to the noise level (2 units standard deviation), so the STA/LTA takes a moment to climb past the threshold. The S-wave, being much larger (300 units amplitude), is detected very reliably and almost immediately when it arrives.

The distance estimate works reasonably well but depends heavily on how accurately the S-P time is measured. Even a 0.5 second error in detection can lead to a distance error of a few kilometers. The magnitude estimate is the roughest result, since it is very sensitive to the amplitude reading and the distance estimate, both of which have some error in them.

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6. Conclusions

This project successfully demonstrates how seismic wave processing works in a step-by-step, understandable way. Starting from raw synthetic data all the way to estimating earthquake magnitude, each stage of the pipeline was implemented and explained.

The key takeaways from this project are:

- Seismic signals carry useful information about earthquakes, and the gap between P and S waves is the most important piece of data for distance estimation.
- Digital signal processing, specifically bandpass filtering, is essential for cleaning up noisy signals before analysis.
- The STA/LTA algorithm is a simple but effective method for detecting wave arrivals automatically. It is still widely used in modern seismic networks.
- Distance can be estimated from a single station, but the accuracy depends on how precise the wave detection is.
- Richter magnitude estimation is possible using the amplitude of the S-wave and the distance, though it is inherently an approximation when done this simply.

Overall, the simulation behaves as expected and the results are physically reasonable. The approach taken here is a simplified but faithful representation of what real seismic processing software does.

7. Limitations

While the program works well for its purpose, there are several important limitations that should be acknowledged:

1. Synthetic Data Only: The signal is artificially generated and does not capture the full complexity of real earthquake waveforms. Real seismograms can have many more overlapping phases, ground resonance effects, and site-specific noise patterns that this simple model cannot replicate.

2. Single Station: A real earthquake location requires data from at least three seismograph stations. With one station you can only get a distance estimate (a circle), not an actual location. Adding more stations and using triangulation would make the system much more useful.

3. Fixed Wave Velocities: The P and S wave speeds (6 km/s and 3.5 km/s) were assumed to be constant everywhere. In reality, wave speeds change depending on the type of rock the wave is passing through. Using a more detailed Earth model would improve distance accuracy.

4. Simplified Magnitude Formula: The Richter magnitude formula used here is an approximation. Real magnitude calculations use more sophisticated methods and calibrations that vary depending on the region and the specific seismograph being used.

5. No Depth Consideration: This program assumes the earthquake happened on the surface, but in reality earthquakes can occur at depths of anywhere from a few kilometers to several hundred kilometers. Depth affects both the wave travel times and the observed amplitudes.

6. Fixed Threshold: The STA/LTA threshold of 3 was set manually. In real systems, adaptive thresholds are used that adjust based on the local noise level. A fixed threshold can miss small earthquakes or produce false detections during noisy periods.

7. Same random noise seed each run: Because a new random noise is generated every time the code runs, the results are slightly different each time. This would not be an issue with real recorded data but makes it slightly harder to reproduce exact numbers from this simulation.

Done by :
Veer Nagpal (241EC161)
Shouryadip C (241EC152)